

Dr. Sven Möller

Assistant Professor

Personal Information

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Employment

- from 2026 **Assistant professor (lecturer), tenured**, *Maynooth University*, Ireland
- 2022–2026 **Junior research group leader**, *Universität Hamburg*, Germany
Emmy Noether Programme: *Vertex algebras associated with 2- and 4-dimensional conformal field theories* (positive interim evaluation, 2025)
Principal investigator in the Collaborative Research Cluster (SFB 1624): *Higher Structures, Moduli Spaces and Integrability*
Key researcher in Cluster of Excellence *Quantum Universe*
- (2024) **Visiting professor (Vertretungsprofessur)**, *Technische Universität Darmstadt*, Germany
- 2020–2021 **Postdoctoral research fellow**, *Research Institute for Mathematical Sciences*, Kyoto, Japan
JSPS project: *Construction and classification of vertex operator algebras related to 2- and 4-dimensional conformal field theories*
- 2017–2020 **Assistant professor, non-tenure track**, *Rutgers University*, USA
- 2013–2017 **Doctoral research associate**, *Technische Universität Darmstadt*, Germany
DFG project: *Infinite-dimensional Lie algebras in string theory*
- 2010–2013 **Teaching assistant**, *Technische Universität Darmstadt*, Germany

Education

- 2013–2016 **Ph.D. Mathematics**, *Technische Universität Darmstadt*, Germany, with distinction
Thesis: “A Cyclic Orbifold Theory for Holomorphic Vertex Operator Algebras and Applications” (Advisor: Nils Scheithauer)
- 2010–2012 **M.Sc. Mathematics**, *Technische Universität Darmstadt*, Germany, with distinction
Thesis: “On the Classification of Automorphic Products of Singular Weight”
- 2010–2012 **M.Sc. Physics**, *Technische Universität Darmstadt*, Germany, with distinction
Thesis: “Pion-Pion Scattering and Shear Viscosity in the Nambu–Jona-Lasinio Model”
- 2009–2010 **Abroad studies**, *Lund University*, Sweden
- 2007–2010 **B.Sc. Mathematics**, *Technische Universität Darmstadt*, Germany, with distinction
- 2007–2010 **B.Sc. Physics**, *Technische Universität Darmstadt*, Germany, with distinction

Grants, Scholarships and Awards

- 2026–2030 **Grant**, *ARDÚ Doctoral Scholarship Funding*, Maynooth University, €100,000

- 2024–2027 **Research grant**, Collaborative Research Cluster (SFB 1624), Deutsche Forschungsgemeinschaft (German Research Foundation), € 10 mil. (one of 16 PIs)
- 2022–2028 **Research grant**, Emmy Noether Programme, Deutsche Forschungsgemeinschaft (German Research Foundation), € 1,195,200 (sole PI)
- 2020–2023 **Research grant**, Grant-in-Aid KAKENHI 20F40018, Japan Society for the Promotion of Science (JSPS), ¥ 1,500,000 (~€ 12,000)
- 2020–2022 **Fellowship**, Postdoctoral Fellowship for Research in Japan, Japan Society for the Promotion of Science (JSPS), ¥ 8,888,000 (~€ 70,000)
- 2018–2020 **Grant**, AMS-Simons Travel Grant, American Mathematical Society (AMS), \$ 4,000 (~€ 3,500)
- 2013–2016 **Doctoral scholarship**, Studienstiftung des deutschen Volkes (German Academic Scholarship Foundation)
- 2013 **Doctoral scholarship**, Stiftung der Deutschen Wirtschaft (Foundation of the German Economy), declined
- 2011 **B.Sc.-Prize**, Gerhard-Herzberg-Gesellschaft
- 2007–2012 **Scholarship**, Studienstiftung des deutschen Volkes (German Academic Scholarship Foundation)

Selected Publications

- [MS23] Sven Möller and Nils R. Scheithauer. Dimension formulae and generalised deep holes of the Leech lattice vertex operator algebra. *Ann. of Math.*, 197(1):221–288, 2023. (arXiv:1910.04947 [math.QA]).
- [AKM23] Tomoyuki Arakawa, Toshiro Kuwabara and Sven Möller. Hilbert schemes of points in the plane and quasi-lisse vertex algebras with $\mathcal{N} = 4$ symmetry. 2023, submitted. (arXiv:2309.17308 [math.RT]).
- [HM23] Gerald Höhn and Sven Möller. Classification of self-dual vertex operator superalgebras of central charge at most 24. *Selecta Math.*, to appear, 2026. (arXiv:2303.17190 [math.QA]).
- [GLM24b] César Galindo, Simon Lentner and Sven Möller. Modular $\mathbb{Z}_2$ -crossed Tambara-Yamagami-like categories for even groups. *Adv. Math.*, 493:110902, 2026. (arXiv:2411.12251 [math.QA]).
- [EMS20a] Jethro van Ekeren, Sven Möller and Nils R. Scheithauer. Construction and classification of holomorphic vertex operator algebras. *J. Reine Angew. Math.*, 759:61–99, 2020. (arXiv:1507.08142 [math.RT]).

All Publications

- [AFM26] Tomoyuki Arakawa, Andrea E. V. Ferrari and Sven Möller. Vertex Superalgebras for Hypertoric Varieties and 3d Abelian Gauge Theories. 2026, submitted. (arXiv:2606.24708 [math.QA]).
- [AM25] Dražen Adamović and Sven Möller. Character Identities Between Affine and Virasoro Vertex Operator Algebra Modules. *Math. Z.*, to appear, 2026. (arXiv:2511.20121 [math.QA]).

- [GLM24b] César Galindo, Simon Lentner and Sven Möller. Modular $\mathbb{Z}_2$ -crossed Tambara-Yamagami-like categories for even groups. *Adv. Math.*, 493:110902, 2026. (arXiv:2411.12251 [math.QA]).
- [MR24b] Sven Möller and Brandon C. Rayhaun. Equivalence relations on vertex operator algebras, II: Witt equivalence and orbifolds. *Ann. Henri Poincaré*, to appear, 2026. (arXiv:2410.18166 [hep-th]).
- [GLM24a] César Galindo, Simon Lentner and Sven Möller. Computing G -crossed extensions and orbifolds of vertex operator algebras. 2024, submitted. (arXiv:2409.16357 [math.QA]).
- [MR24a] Sven Möller and Brandon C. Rayhaun. Equivalence relations on vertex operator algebras, I: Genus. 2024, submitted. (arXiv:2408.07117 [hep-th]).
- [AKM23] Tomoyuki Arakawa, Toshiro Kuwabara and Sven Möller. Hilbert schemes of points in the plane and quasi-lisse vertex algebras with $\mathcal{N} = 4$ Symmetry. 2023, submitted. (arXiv:2309.17308 [math.RT]).
- [HM23] Gerald Höhn and Sven Möller. Classification of self-dual vertex operator superalgebras of central charge at most 24. *Selecta Math.*, to appear, 2026. (arXiv:2303.17190 [math.QA]).
- [MS21] Sven Möller and Nils R. Scheithauer. A geometric classification of the holomorphic vertex operator algebras of central charge 24. *Algebra Number Theory*, 18(10):1891–1922, 2024. (arXiv:2112.12291 [math.QA]).
- [HM22] Gerald Höhn and Sven Möller. Systematic orbifold constructions of Schellekens’ vertex operator algebras from Niemeier lattices. *J. London Math. Soc.*, 106(4):3162–3207, 2022. (arXiv:2010.00849 [math.QA]).
- [ELMS21] Jethro van Ekeren, Ching Hung Lam, Sven Möller and Hiroki Shimakura. Schellekens’ list and the very strange formula. *Adv. Math.*, 380:107567, 2021. (arXiv:2005.12248 [math.QA]).
- [MS23] Sven Möller and Nils R. Scheithauer. Dimension formulae and generalised deep holes of the Leech lattice vertex operator algebra. *Ann. of Math.*, 197(1):221–288, 2023. (arXiv:1910.04947 [math.QA]).
- [Möl21] Sven Möller. Natural construction of ten Borcherds-Kac-Moody algebras associated with elements in M_{23} . *Comm. Math. Phys.*, 383(1):35–70, 2021. (arXiv:1905.09629 [math.QA]).
- [Möl18] Sven Möller. Orbifold vertex operator algebras and the positivity condition. In *Research on algebraic combinatorics and representation theory of finite groups and vertex operator algebras*, number 2086 in RIMS Kôkyûroku, pages 163–171. Research Institute for Mathematical Sciences, 2018. (arXiv:1803.03702 [math.QA]).
- [EMS20b] Jethro van Ekeren, Sven Möller and Nils R. Scheithauer. Dimension formulae in genus zero and uniqueness of vertex operator algebras. *Int. Math. Res. Not.*, 2020(7):2145–2204, 2020. (arXiv:1704.00478 [math.QA]).

- [Möl16] Sven Möller. *A cyclic orbifold theory for holomorphic vertex operator algebras and applications*. Dissertation, Technische Universität Darmstadt, 2016. (arXiv:1611.09843 [math.QA]).
- [EMS20a] Jethro van Ekeren, Sven Möller and Nils R. Scheithauer. Construction and classification of holomorphic vertex operator algebras. *J. Reine Angew. Math.*, 759:61–99, 2020. (arXiv:1507.08142 [math.RT]).

Research Visits

- Mar. 2025 Simons Center for Geometry and Physics programme *Supersymmetric Quantum Field Theories, Vertex Operator Algebras, and Geometry*, Stony Brook, USA
- Nov. 2024 Tomoyuki Arakawa, Research Institute for Mathematical Sciences, Kyoto, Japan
- Mar. 2024 Tomoyuki Arakawa, Ningbo University, China
- Oct. 2023 Dražen Adamović, University of Zagreb, Croatia
- Nov. 2022 Gerald Höhn, Kansas State University, USA
- Oct. 2022 CRM thematic month *Quantum symmetries: tensor categories, topological quantum field theories, vertex algebras*, Montreal, Canada
- Mar. 2022 IMPA thematic programme *Vertex and chiral algebras*, Rio de Janeiro, Brazil
- June 2019 Nils Scheithauer, Technische Universität Darmstadt, Germany
- Nov. 2018 Gerald Höhn, Kansas State University, USA
- Aug. 2018 Ching Hung Lam, Academia Sinica, Taiwan
- July 2018 RIMS research project *Vertex operator algebras and symmetries*, Kyoto, Japan
- June 2018 Nils Scheithauer, Technische Universität Darmstadt, Germany
- Sept. 2017 Gerald Höhn, Kansas State University, USA

Invited Talks and Conferences

- July 2026 *Topics in Theoretical and Mathematical Physics*, Universität Mainz, Germany
- Mar. 2026 *Vertex Algebras and Related Topics 2026*, Research Institute for Mathematical Sciences, Kyoto, Japan
- Dec. 2025 *VOAs, Number Theory, and Combinatorics*, Research Institute for Mathematical Sciences, Kyoto, Japan
- Sept. 2025 *International Workshop on Automorphic Forms*, SwissMAP Research Station, Les Diablerets, Switzerland
- July 2025 *Subfactors and Applications*, Mathematisches Forschungsinstitut Oberwolfach, Germany
- June 2025 *Representation Theory XIX*, Dubrovnik, Croatia
- Apr. 2025 *Symplectic Singularities, Supersymmetric QFT, and Geometric Representation Theory*, Simons Center for Geometry and Physics, Stony Brook, USA
- Feb. 2025 *Winter Seminar of the Darmstadt Algebra Group*, Manigod, France
- Dec. 2024 *Algebra and Number Theory Seminar*, Universität zu Köln, Germany
- Nov. 2024 *Representation Theory Seminar*, Research Institute for Mathematical Sciences, Kyoto, Japan

Sept. 2024 *Vertex Algebras and Related Topics*, Research Institute for Mathematical Sciences, Kyoto, Japan
 July 2024 *Joint Meeting AMS/UMI*, Università degli Studi di Palermo, Italy
 July 2024 *Oberseminar Topologie*, Ruhr-Universität Bochum, Germany
 June 2024 *Algebra and Representation Theory in Germany*, RWTH Aachen, Germany
 June 2024 *Vertex Algebras, Geometric Representation Theory and Quantum Groups*, CIRM, Luminy, France
 Mar. 2024 *Workshop on New Directions in Conformal Field Theory*, Universität Hamburg, Germany
 Mar. 2024 *Yongjiang Mathematics Forum Lecture*, Ningbo University, China
 Feb. 2024 *Algebra Seminar*, Technische Universität Darmstadt, Germany
 Nov. 2023 *Algebraic Geometry Seminar*, Stony Brook University, USA
 Sept. 2023 *Komaba Finite Group Theory Seminar*, University of Tokyo, Japan
 Sept. 2023 *iTHEMS Mathematics Seminar*, Riken, Japan
 June 2023 *Quantum Matter Seminar*, Harvard University, USA
 Mar. 2023 *Mini Workshop on Vertex Algebras and Applications*, Zagreb, Croatia
 Mar. 2023 *Winter Seminar of the Darmstadt Algebra Group*, Manigod, France
 Dec. 2022 *Conference in Finite Groups and Vertex Algebras*, Academia Sinica, Taiwan
 Nov. 2022 *Lie Group/Quantum Mathematics Seminar*, Rutgers University, USA
 Oct. 2022 *Algebra Seminar*, Kansas State University, USA
 Oct. 2022 *Representation Theory XVII*, Dubrovnik, Croatia
 Aug. 2022 *Modular Forms in Number Theory and Beyond*, Universität Bielefeld, Germany
 June 2022 *Vertex Algebras and Representation Theory*, CIRM, Luminy, France
 Mar. 2022 *International Workshop on Representation Theory, Vertex and Chiral Algebras*, IMPA, Rio de Janeiro, Brazil
 Mar. 2022 *Lie Group/Quantum Mathematics Seminar*, Rutgers University, USA
 Sept. 2021 *Conference on Vertex Algebras and Related Topics*, Technische Universität Darmstadt, Germany
 Sept. 2021 *MSJ Autumn Meeting 2021*, Chiba University, Japan
 July 2021 *Japan Europe Number Theory Exchange Seminar*, Nagoya University, Japan
 Apr. 2021 *Rocky Mountain Representation Theory Seminar*, University of Colorado, Boulder, USA
 June 2020 *International Seminar on Automorphic Forms*, ETH Zürich, Switzerland
 June 2020 *Oberseminar Geometrische Analysis und Zahlentheorie*, Universität Paderborn, Germany
 Feb. 2020 *Algebra and Number Theory Seminar*, University of Arizona, USA
 Oct. 2019 *AMS Sectional Meeting*, Binghamton University, USA
 Aug. 2019 *Workshop on Vertex Operator Algebras and Related Topics*, Chengdu, China
 June 2019 *Representation Theory XVI*, Dubrovnik, Croatia
 June 2019 *The Mathematical Foundations of Conformal Field Theory and Related Topics*, Chern Institute of Mathematics, Tianjin, China

- Feb. 2019 *Winter Seminar of the Darmstadt Algebra Group*, Manigod, France
- Nov. 2018 *Workshop on Vertex Algebras and Infinite-Dimensional Lie Algebras*, University of Split, Croatia
- Nov. 2018 *Algebra Seminar*, Kansas State University, USA
- July 2018 Participation: *Vertex Operator Algebras and Conformal Field Theory/Vertex Operator Algebras and Symmetries*, Research Institute for Mathematical Sciences, Kyoto, Japan
- June 2018 *Vertex Operator Algebras, Number Theory, and Related Topics: A Conference in Honor of Geoffrey Mason*, California State University, Sacramento, USA
- Apr. 2018 *10th Seminar on Conformal Field Theory: A Conference on Vertex Operator Algebras and Related Topics*, Research Institute for Mathematical Sciences, Kyoto, Japan
- Feb. 2018 *Winter Seminar of the Darmstadt Algebra Group*, Manigod, France
- Dec. 2017 *Research on Algebraic Combinatorics and Representation Theory of Finite Groups and Vertex Operator Algebras*, Research Institute for Mathematical Sciences, Kyoto, Japan
- Sept. 2017 *Algebra Seminar*, Kansas State University, USA
- Jan. 2017 Participation: *Geometry and Representation Theory*, Erwin Schrödinger International Institute for Mathematical Physics, Vienna, Austria
- Sept. 2016 *Lie Group/Quantum Mathematics Seminar*, Rutgers University, USA
- June 2015 *4th Seminar on Conformal Field Theory*, Technische Universität Darmstadt, Germany
- Mar. 2015 Participation: *Characters of Representations and Modular Forms*, Max Planck Institute for Mathematics, Bonn, Germany
- June 2014 *Algebras, Groups and Geometries*, University of Tokyo, Japan
- May 2014 *Algebra & Representation Theory for Young Researchers*, University of Erlangen-Nuremberg, Germany

Colloquium Talks

- Mar. 2026 *Boole Colloquium in Mathematics*, University College Cork, Ireland
- Feb. 2026 *Department of Mathematics and Statistics Colloquia*, University College Cork, Ireland
- May 2024 *Mathematical Colloquium*, Technische Universität Darmstadt, Germany
- July 2022 *Colloquium Mathematical Physics*, Universität Bielefeld, Germany
- May 2022 *Colloquium on Pure Mathematics*, Universität Hamburg, Germany
- May 2022 *Colloquium of the Croatian Mathematical Society*, University of Zagreb, Croatia

Teaching Experience

Lectures

- Spring 2026 Summer school: mathematical aspects of conformal field theories
- Spring 2026 Numerical Analysis
- Fall 2025 Preparatory course: master mathematics
- Spring 2024 Mathematics for computer science II
- Spring 2023 Algebraic geometry

Fall 2022 Preparatory course: master mathematics
 Spring 2022 Summer school: modular forms, vertex algebras and moonshine
 Spring 2021 Modular forms
 Spring 2020 Introductory linear algebra
 Fall 2019 Theory of numbers
 Fall 2019 Introduction to real analysis
 Spring 2019 Introduction to mathematical reasoning (2 x)
 Fall 2018 Introduction to mathematical reasoning
 Spring 2018 Introduction to mathematical reasoning (2 x)
 Fall 2017 Linear algebra
 Spring 2016 Riemann surfaces

Seminars

Spring 2024 Elliptic curves
 Fall 2022 Vertex algebras from twisted QFTs in 3d and 4d
 Fall 2015 Research seminar Darmstadt-Frankfurt: vertex algebras

Research Associate

Spring 2017 Linear algebra II
 Fall 2016 Linear algebra I
 Fall 2015 Linear algebra for physicists I
 Spring 2015 Linear algebra II

Teaching Assistant

2010–2013 Numerous classes in the departments of mathematics and physics

Supervision

since 2025 Postdoc (Thibault Juillard)
 since 2024 Postdoc (Christopher Raymond)
 since 2025 Ph.D. student (Felipe Ruiz)
 2022–2026 Ph.D. student (Hannes Knötzele)
 Two Master's students
 One Bachelor's student

Conferences and Seminars Organised

2026 Co-organiser of *Workshop on Vertex Algebras, Tensor Categories and the Yang-Baxter Equation*, Maynooth University, Ireland
 2022 Co-organiser of *Centre for Mathematical Physics Seminar*, Universität Hamburg and DESY, Germany
 2022 Co-organiser of *Workshop on Modular Forms and Representation Theory*, Manigod, France
 2018 Co-organiser of *Postdoc Colloquium*, Rutgers University, USA
 2017 Co-organiser of *Winter Seminar of the Darmstadt algebra group*, Manigod, France
 2016 Co-organiser of *Winter Seminar of the Darmstadt algebra group*, La Plagne, France

- 2015 Co-organiser of *Summer School on Drinfeld modular forms*, Manigod, France
 2015 Co-organiser of *Winter Seminar of the Darmstadt algebra group*, Manigod, France

Service

- 2025 Member of tenure committee, Universität Hamburg
 2024 Member of hiring committee (W2/W3), Universität Hamburg
 2022 Member of hiring committee (W3), Universität Hamburg
 Referee for Comm. Math. Phys., Transform. Groups, RIMS Kôkyûroku Bessatsu, Math. Z., J. Alg., J. London Math. Soc., Ann. Henri Poincaré, Adv. Math., BIRS Workshops, Invent. Math., Bull. Inst. Math. Acad. Sin., Commun. Contemp. Math., Proc. London Math. Soc., Lett. Math. Phys., Proc. Amer. Math. Soc., SIGMA, Ann. Sci. Éc. Norm. Supér., J. Reine Angew. Math.
 Reviewer for zbMATH, MathSciNet
 since 2014 Member of the selection committee on admission seminars of the *Studienstiftung des deutschen Volkes* (German Academic Scholarship Foundation)

Outreach

- 2021 JSPS Science Dialogue (at secondary school in Fukui Prefecture, Japan)

Memberships

- IMS Irish Mathematical Society
 DMV German Mathematical Society

Languages

German	mother tongue	Norwegian	intermediate
English	fluent	Danish	beginner
Swedish	fluent	Russian	beginner
French	fluent	Japanese	beginner
Latin	knowledge		